

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVPE506G - MATERIAL, TESTING AND EVALUATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Following is **NOT** the thermal property of material:
(a) Temperature (b) Specific Heat
(c) Thermal Conductivity (d) Stress
- 2 Following **IS NOT** the property of Cement:
(a) Fineness (b) Soundness
(c) Setting time (d) Tensile Strength
- 3 RCC does not consists-
(a) Water (b) Admixtures
(c) Retarder (d) Lime
- 4 Following **IS NOT** the standard grade of steel:
(a) Fe415 (b) Fe250 (c) Fe500 (d) Fe275
- 5 _____ is commonly used construction material known for its strength and durability.
(a) Paper (b) Concrete (c) Rubber (d) Cotton
- 6 _____ is used for Glazing in building.
(a) Glass (b) Timber (c) Asphalt (d) Bricks
- 7 Structural insulated panels often use _____ for insulation.
(a) Sand (b) Cement (c) Foam (d) Glass
- 8 Identify the material known for its regenerative crack healing ability.
(a) Zero cement concrete (b) Self-healing concrete
(c) Hemp lime (d) Photo catalytic admixture
- 9 Select the sustainable material derived from plant fibers.
(a) Wood-glass epoxy (b) Bamboo
(c) Photo catalytic admixture (d) Zero cement concrete
- 10 Determine the type of destructive test involving repeated loading
(a) Static testing (b) Impact testing
(c) Cyclic testing (d) Penetration testing
- 11 Select the machine used for compression testing
(a) UTM (b) CTM (c) Hardness tester (d) Loading frame

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- Q2. Solve the following.
- A) State and explain basic properties of engineering materials. 6
 B) Give importance of materials in civil engineering. 6
- Q3. Solve the following.
- A) Explain the manufacturing process of cement. 6
 B) Explain mechanical properties of steel. 6
- Q4. Solve Any Two of the following.
- A) State Difference between RCC & FRC. 6
 B) Explain steel-concrete composite bridge decks. Also state types of the same. 6
 C) Explain Zero Cement Concrete. 6
- Q5. Solve Any Two of the following.
- A) Explain 3D Printing. 6
 B) State the importance of new techniques in construction industry. 6
 C) Explain Self Healing Concrete. 6
- Q6. Solve Any Two of the following.
- A) State the necessity of material testing. 6
 B) State the difference between Destructive Testings and Non-Destructive Testings. 6
 C) Explain UTM- Universal Testing Machines with neat sketch. 6

*** End ***